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Analyses of Topical Policy Issues

Do the E-commerce-induced business clusters increase the income of local rural households?

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ABSTRACT

In recent years, rural E-commerce clusters, represented by Taobao Villages, have emerged rapidly in China, becoming a significant force influencing regional economic development. To explore the impact of the formation of Taobao Villages on the household income of local residents, this paper employs a Difference-in-Differences (DID) approach and conducts an analysis based on data from the China Family Panel Studies (CFPS) from 2010 to 2020. The empirical findings show that: Firstly, on average, the formation of Taobao Villages negatively impacts local household income, resulting in a 5.3% decrease. Secondly, the formation of Taobao Villages boost local business income but reduce migrant workers' wage income by drawing them back from higher-paying jobs. Collectively, the opportunity cost of forgone wages outweighs the increase in business income. Finally, heterogeneity analysis reveals that this impact is uneven across different groups, with the formation of Taobao Villages exacerbating income inequality among rural households in China. These findings suggest that policymakers should adopt a cautious approach to promoting E-commerce-induced business clusters, prioritizing equity safeguards to mitigate risks of declining labor income and unequal benefits among the population.

1. Introduction

E-commerce development has brought more business opportunities to the economy by reducing the cost of accessing potential consumers and logistics costs. This is particularly significant for households living in rural areas, with a higher cost of selling their advantaged products and services than their urban counterparts with a more advanced market economy. One well-documented feature of the E-commerce development in rural economies is the formation of business clusters. After the first batch of successful entrepreneurs gain profits from E-commerce, more entrants start their businesses in similar or relevant types of businesses. This sort of chain effect can be explained by various mechanisms, including social networks, imitation behaviors, knowledge spillover, and competition. It consequently fosters the formation of business clusters with a group of commercial units in one or more relevant industries (Gill et al., 2007; Glaeser et al., 2010; Zeng et al., 2019). These mechanisms, in turn, contribute to the formation of business clusters induced by E-commerce (henceforth, E-business clusters), which are expected to influence household income in various ways. The government also considers them a tool to improve rural residents' welfare.[1] Theoretically, for farmers, the E-commerce platform facilitates the sales of goods, directly increasing their incomes from selling agricultural products (Shimamoto

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[1] For instance, the Chinese central government announced a plan to support the development of E-commerce in rural areas in March 2024 and considers it an efficient way to improve the income of local farmers.

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et al., 2015; Luo and Niu, 2019b). Moreover, the formation of E-business clusters has spillover effects, including the upgrading of different industries (Li and Qin, 2022; Zhang et al., 2024) and knowledge spread across industries (Lin, 2019; Yang, 2022). These spillover effects further improve the local residents' income in general.

In this study, we propose an additional perspective to explain the impact of the E-business clusters by considering (temporary) migration between urban and rural areas. Given limited job opportunities locally, some members of rural households choose to be temporary migrant workers, i.e., work in cities for a higher income to support their families. The formation of E-business clusters in their hometowns offers more local job and business opportunities (Koengkan et al., 2024). This would further incentivize some migrant workers who used to work outside to return to their hometowns for non-pecuniary causes, such as lower costs of looking after their children and parents. From a pecuniary perspective, some migrant workers may return to their hometowns because of a better job market due to the E-business clusters. This can reduce the net income gap between rural and urban jobs, considering the expensive living costs in cities. As a result, these migrant workers could return to their hometowns to start businesses or be employed by local enterprises even though their income is lower than before. Consequently, the direction of the impact of the E-business clusters on local households' income remains unclear, and the income structure could change. Empirical evidence on this issue is scarce as current literature mainly focuses on the formation, agglomeration, and development patterns of the E-business clusters using theoretical models or case studies (Qi et al., 2019; Wei et al., 2020; Liu et al., 2020; Zang et al., 2023). Both Ye et al. (2020) and Mei et al. (2020) find that households in areas with E-business clusters are more likely to be entrepreneurs (Ye et al., 2020; Mei et al., 2020). The development of E-business clusters and the relevant business can also boost regional economic growth by attracting more laborers, particularly those who used to be migrant workers (Lin, 2019; Liu, 2020). Although these findings evaluate how E-business clusters affect local households' income to some extent, the direct effect on local households' income remains unclear (Tang and Zhu, 2020; Tang et al., 2022).

Our study examines the impact of E-business clusters on local farmers' income in China. There are several reasons for us to focus on the Chinese setting. First, the burgeoning development of E-commerce in China not only makes the country the largest E-commerce market but also facilitates the formation of relevant local business clusters. Alibaba Group, a conglomerate specializing in E-commerce and one of the largest players in the market in China, has released the list of villages or towns where the local online business has become "clustered" according to multiple criteria such as online transaction volume. These villages are called "Taobao Villages", and such villages have reached 7023, approximately accounting for 1% of the total number of administrative villages nationwide. Second, a large population in less developed areas, particularly villages, are migrant workers outside their hometowns for a higher wage or income. When they are encouraged by the development of E-business at home and return to their homes, the opportunity cost cannot be ignored when evaluating the effect of E-business. Last but not least, E-business clusters are packed with small enterprises or even individuals from rural areas in China; therefore, such development is considered a potentially effective way to raise the income of households in these peripheral areas. Understanding the effect of E-business clusters on local residents' income can also be helpful for other developing countries.

In this study, we use Chinese Family Panel Studies (CFPS) data from 2010 to 2020 to examine the effects of E-business clusters, measured by the emergence of Taobao Villages, on local households' income. In contrast to most studies focusing on the effect at the macro economy level, we find that the Taobao Villages exhibit a negative effect on overall income at the local household level. Further analysis shows that Taobao Village does increase the local households' business income but also decreases their wage incomes, resulting in an overall decline in income. The decrease in wages for migrants who used to work outside their hometowns for higher wages and who returned to be local workers encouraged by the rise of Taobao Villages can be attributed to our finding of the decrease in households' wage incomes. Moreover, the switch from the employed to entrepreneurs indicates a structural change in the overall income composition, which can also explain the decrease in wage income.

This study contributes to the literature on the economic consequences of the development of E-commerce by providing new empirical evidence. While most current studies have found a positive effect of E-commerce on income, we find a negative impact of E-business clusters on local households' income. We attribute the adverse effect to the opportunity cost of being entrepreneurs encouraged by local E-business clusters, particularly for migrant workers who used to earn higher wages outside their hometowns (Qi et al., 2019; Li et al., 2021). These results and discussions contribute to providing a new analytical framework and evidence for how digital technology influences population mobility and income. We additionally uncover the heterogeneous effects of E-business clusters on income and the increasing income inequality in the region after the formation of E-business clusters (He et al., 2024). Beyond various mechanisms through which E-commerce benefits incomes (such as mitigating information asymmetry in Zhou et al., 2021 and Li and Qin, 2022), our heterogeneity helps better understand the sub-population who benefit more. Furthermore, we investigate the impact of E-business clusters on incomes at the household level, supplementing the empirical evidence to the current literature, which mostly focuses on macro-level impacts.

Our study also provides insight into understanding the socioeconomic impact of E-commerce development. China's E-business cluster experience provides a cautionary template for digitizing economies. The observed shifts — from wage labor to business income and altered migration patterns — mirror challenges in regions like Southeast Asia and Latin America, where the E-commerce platform growth disrupts informal labor markets. While clusters can spur entrepreneurship, they risk exacerbating inequality in contexts with weak safety nets. Notably, the net income decline (driven by high wage opportunity costs) for local households in the Chinese context may contrast sharply with lower-wage economies like Sub-Saharan Africa, where digital self-employment could benefit households. These insights urge policymakers to tailor safeguards to local wage levels and institutional capacities.

The remaining sections of this paper are structured as follows: Section 2 presents the background. Section 3 presents a systematic literature review. Section 4 shows a simple theoretical model. Section 5 describes the data. Section 6 presents the empirical estimation strategy and estimation results. Section 7 explores mechanisms. Section 8 offers discussions on heterogeneity. Section 9 concludes with implications.

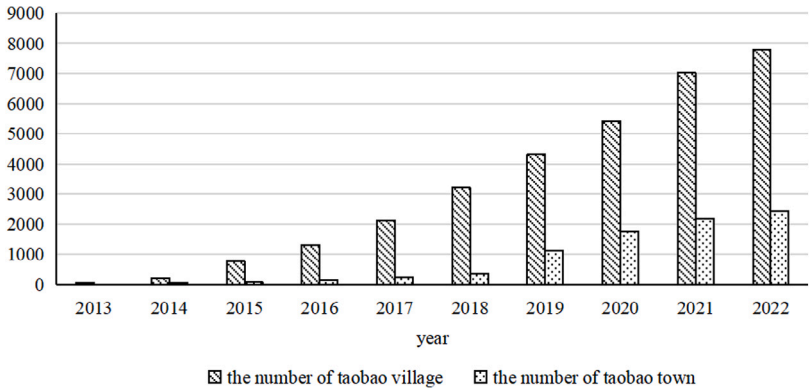

Fig. 1. Growth trend of Taobao village.

2. E-commerce in rural China

The rural E-commerce transaction volume in 2020 reached 1.79 trillion yuan, and it is expected to reach 2.8 trillion yuan by 2025.[2] E-commerce, as a crucial application of digital technology, is helpful to break the temporal and spatial limitations of transactions, attracting people to start E-commerce businesses. After discovering the leaders' success in the village, more and more merchants are imitating and engaging in neighborhood communication, learning the experience and skills of typical cases in a low-cost way, and starting to operate online stores. Subsequently, with the increase of online stores, the chain effect leads to the development of service industries such as express delivery and logistics. Competition and cooperation among online stores reduce entrepreneurial costs, improve entrepreneurial levels, and form an E-commerce entrepreneurial cluster, known as Taobao Village.

A Taobao Village refers to a considerable number of online merchants congregating in a specific village, primarily utilizing Taobao as the main trading platform, and leveraging the Taobao E-commerce ecosystem to establish a scaled and synergistic E-commerce industry cluster. In fact, Taobao Village is not only an academic concept but also fulfills criteria accredited by Alibaba Group's Alibaba Research Institute. Since 2013, this research institute has annually published reports and disclosed the certified lists of Taobao Villages and Taobao Towns on its website. The certification criteria primarily encompass three requirements: firstly, the operational location must be in rural areas, with administrative villages as the basic unit; secondly, the village must have more than 100 active online stores, or more than 10% of local households have active online stores; thirdly, the total annual transaction amount of all online merchants in the village must exceed 10 million yuan. Only when all three criteria are met simultaneously will a village be certified as a Taobao Village.

Taobao Village certification is not merely a label but a threshold indicator for two transformative processes. First, certification requires villages to meet stringent criteria (e.g., ≥100 active online shops and ≥1.5M annual E-commerce revenue), which act as a catalyst for E-business cluster formation. Meeting these criteria triggers structural changes such as economies of scale, supply chain integration, and the emergence of specialized services (e.g., logistics hubs) that are largely absent in non-certified villages. Second, certification brings heightened public exposure and targeted policy support. Certified villages gain visibility on platforms like Alibaba's "Taobao Village" portal, attracting entrepreneurs and customers, especially the local households. Non-certified villages, even those with nascent E-commerce activity, lack access to these resources, creating a clear divergence in economic trajectories.[3]

Once established, Taobao Villages exhibit a phenomenon of explosive growth. As is shown in Fig. 1, by 2021, the number of Taobao Villages reached 7780, accounting for more than 1% of the total number of administrative villages nationwide. As disclosed in the "China Taobao Village Research Report," the sales volume of Taobao Villages (and Towns) reached 220 billion yuan in 2018 and surpassed 1 trillion yuan in 2022. It is evident that an E-commerce revolution is rapidly unfolding in the rural areas of China.

3. Literature review

There are two main strands of literature pertinent to this paper. One focuses on the formation, development, and economic effects of Taobao Villages, while the other explores the relationship between E-commerce and farmers' income.

[2] According to data from the Ministry of Commerce.

[3] Taking Dongfeng Village in Shaji Town, Suining County, Jiangsu Province as an example, Dongfeng Village is hailed as the "first Taobao village in China". Prior to 2006, the local area was widely known as a poverty-stricken village. In 2006, three young individuals in the village, Sun Han, Chen Lei, and Xia Kai, fortuitously discovered business opportunities on the online platform Taobao. Subsequently, they procured simple assembly furniture offline and began selling them through their online stores. Remarkably, in the first month alone, they achieved sales exceeding one million RMB. This initial success prompted a wave of emulation among the surrounding villagers, consequently igniting the local e-commerce industry. Presently, over 3000 households, accounting for 80% of the village's total, have established their own Taobao shops. The furniture output value of the entire Shaji Town has surpassed 10 billion RMB, earning it the accolade of the "Shaji Model". This account exemplifies how a single computer and an internet connection have fostered an entire industry, representing not only the story of Dongfeng Village but also serving as an epitome of thousands of Taobao villages across China.

1. Formation, development and economic effects of Taobao villages

Taobao villages represent rural industrial clusters driven by E-commerce. Existing studies primarily study their formation process and influencing factors through two lenses: entrepreneurial management and cluster evolution. From the entrepreneurial management perspective, research focuses on: (1) the motivations, capabilities, and entrepreneurial behaviors of entrepreneurial groups (Avgerou and Li, 2013; Lin et al., 2016; Luo and Niu, 2019a; Yu and Cui, 2019; Qiao and Qiu, 2024); and (2) the roles of E-commerce platforms, industrial foundations, social networks, and government support in facilitating local residents' E-commerce entrepreneurship and the formation of villages (Qi et al., 2019; Tang and Zhu, 2020; Liu et al., 2020; Wei et al., 2020; Gao et al., 2022). Recent studies also characterize farmers' entrepreneurship in Taobao villages as a representative Bottom of the Pyramid (BoP) innovation model (Gao and Liu, 2020).

The cluster evolution perspective emphasizes holistic cluster dynamics rather than individual entrepreneurs. This stream: (1) divides cluster evolution into germination, growth, and maturity stages while identifying key drivers at each phase (Leong et al., 2016; Zeng et al., 2019); and (2) analyzes internal and external factors in rural E-business cluster evolution using theoretical frameworks such as the diamond model, ecosystem theory, and space of flows (Guo et al., 2014; Cui et al., 2019; Zhang et al., 2022). Subsequent research further explores clusters' socioeconomic impacts, including effects on farmer income, regional development, and ecological environments (Li et al., 2021; Zhang et al., 2024).

2. E-commerce and Farmer Income

The extensive literature demonstrates that E-commerce, as a quintessential digital technology, enhances income growth in rural and remote areas by improving information accessibility, reducing transaction costs, and connecting these regions to broader market (Li et al., 2021; Li and Qin, 2022; Chen et al., 2023; Li and He, 2024). Studies also identify E-commerce's poverty alleviation effects (Li et al., 2019; Peng et al., 2021). Specifically, for Taobao villages, some researchers investigate income effects using county-level panel data (Tang et al., 2022; Li and Qin, 2022; Zhang et al., 2024). Li and Qin (2022) emphasize that Taobao villages drive the transformation of the industrial structure to increase local income, while Zhang et al. (2024) identify three pathways: to foster new business creation, to reduce the operational costs of E-commerce, and to facilitate cross-sector mobility of farmers. Case studies and small-sample surveys further corroborate these findings (Luo and Niu, 2019a; Li et al., 2021; Zhang et al., 2022). In particular, Zhang et al. (2022) document a change in income composition: traditional agricultural production and migrant labor incomes have been replaced by E-commerce revenues as the primary income source after the formation of the Taobao village.

However, some studies present different findings. Couture et al. (2021) question the magnitude of E-commerce's income effects, while Li et al. (2021) reveal a dual impact: positive effects on business income but negative impacts on wage income. Although the latter study leaves this paradox unexplained, our findings align with their results and further demonstrate that wage income reduction stems from residents forgoing higher-paying migrant work opportunities for local employment. Previous literature emphasizes the role of Taobao villages in attracting return migration and external labor inflows (Lin, 2019; Liu et al., 2020), but overlooks the opportunity costs associated with these choices, which is a critical contribution of this study.

Divergent findings between studies may also arise from methodological differences. The aggregate county-level data often do not capture individual behavioral changes, particularly in relation to income structure. Meanwhile, case studies and small-sample surveys frequently focus on exemplary regions, potentially exaggerating Taobao villages' impacts. Differentiating from these approaches, our analysis utilizes nationally representative CFPS data to address sample selection biases, estimate actual effects, and investigate employment and mobility patterns through comprehensive indicators, thereby elucidating underlying mechanisms.

4. Economic model

The idea is to set up a migration decision that considers both monetary return and non-monetary return. We use consumption to represent the former and the number of children for the latter. Why the number of children? First, our prescribed mechanism indicates that taking care of children is one type of non-monetary return. Second, the number of children is positively associated with the cost of child-bearing (e.g., time spent with children). For rural residents, the cost is higher in cities and lower in rural areas. Third, when we want to construct equilibrium, we need to introduce population growth where the number of children is a key element. It is worthy to note that the model endogenizes both consumption and number of children. The number of children can be interpreted as fertility decision and also child-bearing-related decision when assuming the cost/time spent is monotonically associated with the number of children.

For residents in rural areas, household utility is

$$U^R = \ln c + \beta \ln n^R = \ln(w^R - \phi^R n^R) + \beta \ln n^R \tag{1}$$

with technology function $Y^R = A^R N^R$

For residents in urban areas, household utility is

$$U_i^U = \ln c_i + \beta \ln n_i^U = \delta_i = \ln(w^U - \phi^U n_i^U) + \beta \ln n^U - \delta_i \tag{2}$$

with technology function $Y^U = A^U (N^U)^\alpha (\bar{N^U})^{1-\alpha}$ where

- c: consumption
- n: number of children
- ϕ^R, ϕ^U: child-bearing cost in rural (urban)
- δ_i: disutility of staying at urban area

Spatial equilibrium, or locational no-arbitrage condition is

$$\ln(\frac{w^U}{1+\beta}) + \beta \ln \frac{\beta w^U}{(1+\beta)\phi^U} - \delta_i^* = \ln(\frac{w^R}{1+\beta}) + \beta \ln \frac{\beta w^R}{(1+\beta)\phi^R} \tag{3}$$

or

$$\delta_i^* = \ln(\frac{w^U}{w^R}) + \beta \ln \frac{w^U/\phi^U}{w^R/\phi^R} \tag{4}$$

By definition, the migration population M_{t+1} is

$$N_{t+1}^R = N_t^R(1+n^R) - M_{t+1}$$
$$N_{t+1}^U = N_t^U(1+n^U) + M_{t+1}$$

Assuming an uniform distribution of $\delta_i \sim U(0, \bar{D})$, we can also show that

$$M_t = \frac{1}{\bar{D}}(\delta_{i^*,t} - \delta_{i^*,t-1}) = \frac{1}{\bar{D}}[(1+\beta)\ln(1+\gamma_{A,t}) - \beta \ln(1+\gamma_{\phi,t})] \tag{5}$$

where $1+\gamma_{j,t} \equiv \frac{g_{j,t}}{g_{j,t-1}}$ $j \in \{A, \phi\}$, $g_{A,t} = \frac{A^U}{A^R}$, and $g_{\phi,t} = \frac{\phi^U}{\phi^R}$

At Boundedly Generalized Stable Equilibrium (BGSE, one type of equilibrium where n^R/n^U is constant), we have constant growth rates so that $\gamma_{A,t} = \gamma_{\phi,t} = \gamma$. This in turn yields

$$M_t = \frac{1}{\bar{D}} \ln(1+\gamma) > 0 \tag{6}$$

When does internal migration cease?

$$\beta \ln(\phi^U/\phi^R) > (1+\beta)\ln\alpha + (1+\beta)\ln(A^U/A^R) - \delta_{min} \tag{7}$$

5. Data

We use data from the CFPS which is a panel survey collecting economic and demographic information from Chinese households, starting from 2010 and collected every two years. We use five waves of data from 2010 to 2020, including 13,235 households from 1922 villages (communities) in 25 provinces.[4] We also need the list of Taobao Villages to construct the key variable of interest, which is obtained from Alibaba Research Institute. All other variables at macroeconomic (county) level are from the National Bureau of Statistics.

5.1. Outcome variable

The outcome variable is household income directly provided by respondents. We use total income in the baseline regression and per capita income in robustness checks. In the mechanism analysis, we further divide the total income into different sources: (1) wage income, (2) business income, (3) investment income, and (4) transfers. For all measures of income in the regression, we implement 1% winsorization at both ends and add 1 when taking the logarithms.

5.2. Treatment variable

The regional and dynamic development of Taobao villages is of interest to construct our treatment variable. In the baseline specification, we use a binary variable that equals one if the household lives in a county that has Taobao Villages in the given year, and zero otherwise. The Alibaba Research Institute has published the shortlist of Taobao villages every year since 2013, but the CFPS collects data every two years since 2010. We specify that counties having Taobao villages for the first time in 2013, 2015, 2017 and 2019 receive the treatment from the following year, and counties having Taobao villages in 2014, 2016, 2018 and 2020 receive the treatment from the current year.

Such binary treatment differentiates counties without any Taobao Village and counties with at least one Taobao Village. Accordingly, we specify the former group as the establishment of E-business clusters and the latter as the control group. The treatment effect is therefore interpreted as the impact of E-business clusters on the outcome variable at the extensive margin. Henceforth, we use the impact of Taobao Village and the impact of E-business clusters interchangeably. In robustness checks, we implement alternative measures of the treatment variable including the number of Taobao villages in the county and whether the county has Taobao Town (rather than Taobao Village).[5]

[4] CFPS does not cover data of Tibet, Qinghai, Xinjiang, Ningxia, Inner Mongolia, Hainan, Hong Kong, Macau and Taiwan.

[5] The levels of government in China are province, city, county, town, and village. A village has to meet several criteria to be listed as Taobao Village, and a town is listed as Taobao Town based on similar but more restrict criteria. When there is a Taobao Town in a county, there are normally several Taobao Villages.

Table 1
Summary statistics.

Variable	Obs	Mean	S.D.	Min	Max
Outcome variables (in Chinese RMB)					
Household income	61,746	57,823.32	65,676.76	700	405,000
Household business income	63,097	6846.564	16,689.88	0	105,000
Household wage income	64,371	32,791.47	42,023.99	0	218,000
Treatment variable					
Establishment of Taobao Village	64,490	0.044	0.204	0	1
Control variables					
Number of household members	64,490	3.864	1.773	1	9
Household deposits	64,201	34,273.36	80,652.17	0	500,000
Live in urban area	64,149	0.465	0.499	0	1
Age	57,979	49.648	13.755	20	81
Male	57,979	0.525	0.499	0	1
Education	57,133	2.567	1.289	1	6

The value of education: Illiterate = 1, Elementary school = 2, Junior high school = 3, Senior high school = 4, Junior College = 5, College and above = 6 (the proportion of each value from 1 to 6 is 0.258, 0.237, 0.292, 0.141, 0.044, 0.028). All personal characteristics (age, gender and education level) are based on the house head.

5.3. Control variables

We first include household fixed effects controlling for time-invariant household factors that contribute to family income and thus address time-constant selection bias and year fixed effects controlling for time-varying shocks common to all households. We further control for characteristics at levels of both household and house head that could be potential confounders. The former accounts for resource availability and market access, and it includes household size, household deposits, and location of living (rural or urban). The latter group captures human capital and decision-making dynamics, and it consists of age (and age squared), gender, and education level.[6]

In robustness checks, we include additional characteristics of household and individual (value of land asset and marital status) and economic development of counties (GDP per capita, number of enterprises and number of landline telephone subscribers). It is worth noting that these additional variables are only controlled in robustness checks because there is a large number of observations with missing values about these variable. Table 1 shows the summary statistics of all variables.

5.4. Summary statistics

In our sample, the average household income is around 57,823 Yuan with less than four family members on average. More than half of the household income is attributed to wage income, and business income constitutes about 12% of the total household income. In the periods of our sample (2013–2019), around 4.4% of household-year observations are in the counties with Taobao Villages. The head of the household is 50 years old on average, and there are more males in our sample.

6. Empirical implementation and results

The development of Taobao Village introduced variations in the intensity of E-business clusters of regions (counties) across time periods and geographic locations. Given these variations, we apply Difference-in-Difference (DID) strategy to examine the effect of Taobao Villages on local households' income. Such effect can be identified by examining how the income of households in counties that have Taobao Village changes differently from that of households in counties without Taobao Villages after the corresponding villages are listed as Taobao Villages. The DID specification is presented as follows:

$$Y_{ijt} = \beta_0 + \beta_1 TV_{ij} \times Post_{jt} + \theta_i + \tau_j + \mu_t + \gamma X_{ijt} + \epsilon_{ijt} \tag{8}$$

where Y_{ijt} is the logarithm income of household i in county j at year t. TV_{ij} is a dummy indicating whether county j in which household i lives ever has any Taobao Village. $Post_{jt}$ is a period dummy indicating the years after the emergence of the first Taobao Village in county j. The coefficient β_1 on the interaction term is the treatment effect of interest, capturing the effect of Taobao Village on the income of households living in counties having Taobao Villages. θ_i, τ_j, and μ_t are household fixed effects, county fixed effects, and year fixed effects respectively. It is worthy noting that while the canonical DID model includes separate TV and $Post$ terms, these are subsumed by household and year fixed effects in our framework.

The identification of DID strategy relies on two crucial assumptions. The first is parallel trend assumption which requires that the evolution of household income would have been the parallel between counties having Taobao Villages (treated group) and counties

[6] Education level is measured by a categorical variable: 1 (illiterate), 2 (Elementary school), 3 (Junior high school), 4 (Senior high school), 5 (Junior College), 6 (College and above).

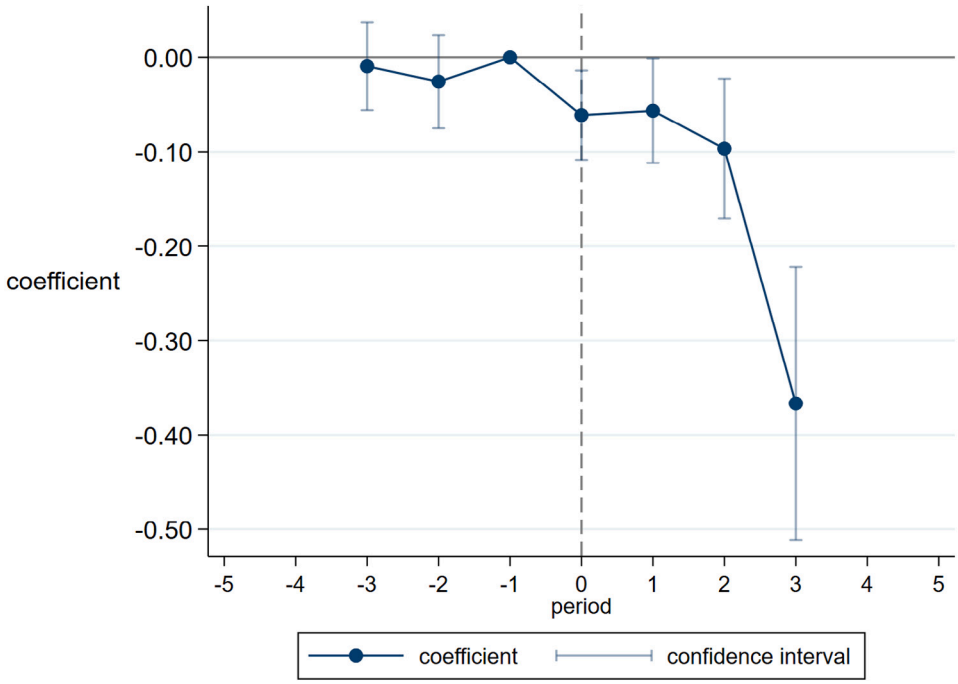

Fig. 2. Regression results from the DID event study. Note: One period prior to the emergence of any Taobao Villages in a county is the baseline group.

without any Taobao Village (control group) in absence of any village listed as Taobao Village. The second is no anticipation effect assumption in which we assume that, in years prior to the first Taobao Village, the household income is not affected by the upcoming emergence of Taobao Village. Equivalently speaking, local households in the county do not know whether and when there will be a village that meets all criteria for being listed as the Taobao Village.

To make our assumptions more reliable, we control for a number of household-specific and region-specific characteristics X_{ijt} that contribute to household income but could have different dynamics between the control and treated group. Precisely, in the baseline specification, we include the number of house members, household deposits, living location (urban or rural), and some characteristics of household heads including age, age squared, gender, and educational background. In robustness checks, we furthermore include county-level factors that capture the macroeconomic development and possible confounding policies.

6.1. Validity of the identification strategy

We start the discussion on the reliability of the assumptions using a routine test showing the difference in household income between the treated and control group before any village is listed as a Taobao Village. We are more confident about the parallel assumption if the difference in the outcome variable between the two groups in the pre-treatment period is trivial. Particularly, we conduct a DID event study regression as follows:

$$Y_{ijt} = \beta_0 + \sum_{k=T_0}^{-1} \beta_k Treat_{ijt} + \sum_{k=0}^{T_1} \beta_k Treat_{ijt} + \theta_i + \mu_t + \gamma X_{ijt} + \epsilon_{ijt} \tag{9}$$

where $T_0 = -3$ and $T_1 = 3$ are the lowest and highest number of lags and leads near the treatment periods, respectively. $Treat_{ijt}$ is a dummy variable, equaling 1 if the observation's period relative to the county j's first period of the emergence of Taobao Villages is k; 0 otherwise (and 0 for observations in villages that are never have any Taobao Village). Fig. 2 summarizes the estimated $\hat{\beta}$ from Eq. (9). The insignificant estimates before the first treatment period indicate that we do not find evidence that household incomes are statistically different between the treated and the control group in the absence of Taobao Village, supporting the parallel assumption.[7]

To further ease the concern that our results are driven by confounding factors other than the development of Taobao Village, we implement a placebo test which relies on the construction of a "fake" treatment group. Precisely, we create a "fake" treatment by randomly selecting counties as treatment units and moving the initial treatment X periods forward. We replicate such procedure 500 times and estimate Eq. (11) as in column (3) of Table 3. If the common support assumption holds, we expect to observe close-to-zero

[7] These analyses suggest that residual confounding, if present, is unlikely to drive our findings.

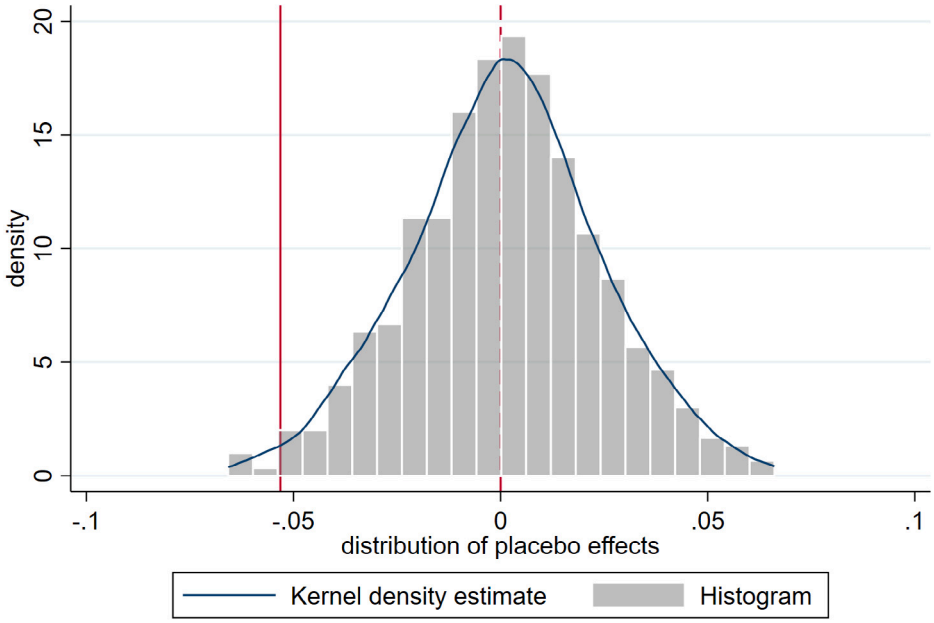

Fig. 3. Distribution of the estimates from placebo test.

estimates when there is no underlying difference in trends between the treated and the control group in the absence of treatment effect we are interested in. Fig. 3 shows the distribution of the DID estimates based on 500 samples. The distribution is close to a normal distribution with zero as the mean. The red vertical line represents the estimate from column (3) in Table 3. We find it is on the left tail and it is statistically different (p-value for the left-sided test: 0.01) from the mean of the estimates from the placebo test. Therefore, our placebo test result provides another evidence supporting the common trend assumption.

No anticipation effect assumes that household income is not affected by the emergence of a Taobao Village in the future. Therefore, we implement a Granger-type causality test as shown in Eq. (10) to find out possible anticipation effect. If households anticipate the coming emergence of a Taobao Village and thus change their behaviors resulting in a different household income prior to the event, we would expect to find out significant α_s which measures the effect of future treatment on current household income. As shown in Table 2, the estimated anticipation effects α_s are statistically insignificant and relatively marginal compared to the contemporaneous effect which are the estimated effect of interest in this study. The statistically insignificant and relatively small estimated α_s are robust to different forward periods ($k \in \{1, 2, 3\}$) included in the specification. As a result, we do not find evidence against the no anticipation effect assumption.

$$Y_{ijt} = \beta_0 + \beta_1 TV_{ij} \times Post_{jt} + \sum_{s=1}^{s=k} \alpha_s TV_{ij} \times Post_{jt+s} + \theta_i + \mu_t + \gamma X_{ijt} + \epsilon_{ijt} \tag{10}$$

6.2. Baseline results

Table 3 presents the impact of Taobao Village on household income. The effect of Taobao Village decreases the local households' income by about 6.3% without controlling any additional variable in X_{ijt}, as shown in column (1). The estimated effect remains stable when controlling for household-specific characteristics whose evolution may be different between the treated and the control group, as shown in column (2). In column (3), we further include individual-specific characteristics of household head to capture the dynamics of the influence of breadwinner on household income. The result is similar to column (1) and (2), suggesting a negative impact of the establishment of Taobao Village. Precisely, the household income decreases by 5.3%, or 3,065 (=5.3%*57,823) Yuan for a household with average income (shown in Table 1).

6.3. Robustness checks

In this section, we check the robustness of the baseline results from Table 3 to alternative specifications. We start by controlling for additional covariates that may have different trends in the treated and control group, which violates the parallel trend assumption. First, besides the family characteristics included in the baseline specification, we include the value of land assets and the marital status of household head. Column (1) of Table 4 shows little change in the result after adding these two additional control variables. Second, because the economic development of regions affects both family income and the development of Taobao Villages, we further include three measures for the macro-economy: GDP per capita in a county, number of industrial enterprises

Table 2
Granger-style causality test results.

	(1)	(2)	(3)	(4)
$TV_{ij} \times Post_{jt}$	−0.053***	−0.067**	−0.028	−0.074
	(0.020)	(0.030)	(0.055)	(0.142)
$TV_{ij} \times Post_{jt+1}$		−0.006	−0.008	0.091
		(0.022)	(0.035)	(0.068)
$TV_{ij} \times Post_{jt+2}$			−0.028	−0.065
			(0.026)	(0.044)
$TV_{ij} \times Post_{jt+3}$				−0.016
				(0.036)
Obs.	53,520	40,461	28,599	18,895

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Standard errors are clustered at household level. Dependent variable is the logarithm of household income. Covariates include number of household members, household deposits, live in urban area, age, age squared, gender, education level, year fixed effects, family fixed effects, and county fixed effects. All individual characteristics are for house head.

Table 3
Effect of Taobao Village on household income with baseline specification.

	(1)	(2)	(3)
$TV_{ij} \times Post_{jt}$	−0.063***	−0.059***	−0.053***
	(0.022)	(0.019)	(0.020)
(log)Number of household members		0.741***	0.719***
		(0.016)	(0.017)
(log)Household deposits		0.025***	0.023***
		(0.001)	(0.001)
Live in urban area		0.071***	0.057**
		(0.024)	(0.026)
Age			−32.982***
			(5.290)
Age squared			15.928***
			(2.578)
Male			0.065***
			(0.012)
Education	N	N	Y
Obs.	61,206	60,707	53,520

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Standard errors are clustered at household level. Dependent variable is the logarithm of household income. Individual characteristics are for the head of house. Education is a categorical variable with the following values: 1 (no education), 2 (primary school), 3 (middle school), 4 (high school or equivalent), 5 (college education), and 6 (Bachelor degree or above). Year fixed effects, family fixed effects, and county fixed effects are included in all regressions.

above designated size, and number of landlines.[8] As shown in column (2) of Table 4, the result still indicates a negative impact of Taobao Village on household incomes at an even larger magnitude. Third, when a village is listed as a Taobao Village, there are not only E-business clusters but also more developed digital infrastructure both of which may affect local households' income. Therefore we add county-level coverage breadth of digital finance.[9] in column (3), and the result still holds. Last, we control for two additional policies that took effect after 2010: comprehensive demonstration of E-commerce entering villages[10] and Informatization in villages[11] The corresponding result is shown in column (4), and the estimate remains robust.

We further check the robustness of our results to alternative specifications. While controlling for the number of household members in the baseline specification, we alternatively use the household income per capita as the outcome variable. Column (1) in Table 5 shows the estimation result, and we find a significantly negative effect that is similar to the baseline result in size. One concern regarding the dichotomous treatment indicator TV_{ij} is that it may not fully capture the heterogeneity in the development of Taobao Villages across counties. Therefore, we use the number of Taobao Villages in a county to replace TV_{ij} in constructing our key interaction term in the regression. An alternative approach is to construct a binary treatment based on the development of Taobao

[8] The number of industrial enterprises above designated size refers to entities with annual main business income of 20 million Yuan or above.

[9] This indicator is one of the dimensions of Peking University Digital Financial Inclusion Index of China, which is measured by the number of Alipay accounts per 10000 people, the proportion of Alipay account binding cards, and the average number of bank cards per Alipay account.

[10] A policy providing financial support for successfully applying regions, aiming to improve the public service system of rural e-commerce, and improve the logistics and distribution system in counties and villages.

[11] This policy requires each pilot province and city to select two counties, ensuring that at least one village-level information service station is built in each administrative village.

Table 4
Robustness check: Additional control variables.

	Baseline (Table 3)	(1)	(2)	(3)	(4)
$TV_{ij} \times Post_{jt}$	−0.053***	−0.049**	−0.096***	−0.094**	−0.085**
	(0.020)	(0.020)	(0.033)	(0.041)	(0.041)
(log)Value of land asset		0.019***	0.023***	0.026***	0.023***
		(0.002)	(0.002)	(0.003)	(0.003)
Married		0.013	0.001	−0.005	−0.040
		(0.022)	(0.038)	(0.043)	(0.048)
(log) GDP per capita			0.181***	0.282***	0.189***
			(0.030)	(0.061)	(0.066)
(log) Number of industrial enterprises above designated size			−0.012	−0.017*	−0.019**
			(0.008)	(0.009)	(0.009)
(log) Number of landlines			−0.015	−0.027	−0.051*
			(0.018)	(0.026)	(0.030)
(log) Coverage breadth of digital finance				0.004	−0.024
				(0.029)	(0.033)
Policy (E-commerce)					0.110***
					(0.028)
Policy (Information)					0.108
					(0.120)
Obs.	53,520	53,173	23,475	16,569	13,733

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Standard errors are clustered at household level. Control variables in all regressions include: (1) family level: number of house members, household deposits, and urban or rural area (2) house head: age, age squared, sex and education level (3) fixed effects: year fixed effects, family fixed effects, and county fixed effects.

Table 5
Robustness check: Alternative specifications and sample.

	Baseline (From Table 3)	(1)	(2)	(3)	(4)	(5)	(6)
$TV_{ij} \times Post_{jt}$	−0.053***	−0.065***			−0.052*	−0.049**	−0.044**
	(0.020)	(0.023)			(0.030)	(0.022)	(0.021)
Number of Taobao Villages			−0.011***				
			(0.004)				
Taobao Town				−0.068***			
				(0.019)			
Sample	Whole	Whole	Whole	Whole	Whole	Whole	Subsample
Outcome	Total	Per capita	Total	Total	Total	Total	Total
Region FE	county	county	county	county	county	village	county
Cluster	Household	Household	Household	Household	Village	Household	Household
Obs.	53,520	53,520	53,520	53,520	53,087	52,414	38,313

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Control variables in all regressions expect for column (1) include: (1) family level: number of house members, household deposits and urban or rural area (2) house head: age, age squared, sex and education level (3) fixed effects: year fixed effects and family fixed effects. In column (2), we do not control number of house members.

Towns rather than Taobao Villages.[12] The estimation results are shown in column (2) and (3) in Table 5. Column (2) indicates that having one more Taobao Village in a county significantly leads to a lower household income by 1.1%. Column (3) shows that the effect of Taobao Town on households' income is significantly negative and slightly larger than that of Taobao Village. Additionally, we show that our results are robust to alternative regional fixed effects and clustering as shown in column (4) and (5). Last, there could be underlying differences between the regions without any Taobao Village (treatment) and others, which threaten the parallel assumption. To alleviate such concern, we drop observations from provinces where there is no Taobao Village. The result based on this subsample is shown in column (6), and the result is still comparable to the one from our baseline specification.

[12] The levels of government in China are province, city, county, town, and village. A village has to meet several criteria to be listed as Taobao Village, and a town is listed as Taobao Town based on similar but more restrict criteria. When there is a Taobao Town in a county, there are normally several Taobao Villages.

Table 6
Robustness check: estimate robust to heterogeneous treatment effects and spillover effect.

	Baseline (From Table 3)	CSDID	Bacon	Spillover
$TV_{ij} \times Post_{jt}$	−0.053***	−0.076***	−0.083***	−0.054**
	(0.020)	(0.028)	(0.028)	(0.021)
Spillover				0.240***
				(0.072)
Obs.	44,446	51,736	20,394	52,318

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Standard errors are clustered at household level. Control variables in all regressions include: (1) family level: number of house members, household deposits and urban or rural area (2) house head: age, age squared, sex and education level (3) fixed effects: year fixed effects, family fixed effects, and county fixed effects.

Additionally, we investigate the interpretation of the estimate when allowing for heterogeneous treatment effects. Given the Two Way Fixed Effects (TWFE) implemented in the previous analysis, the estimated effect should be interpreted as the weighted average of different treatment effects. However, these estimated weights could be negative which deviate the interpretation of our estimate from the true parameter of interest (Baker et al., 2022). To verify the interpretation of our result, we follow the method robust to heterogeneous treatment effects by Callaway and Sant'Anna (2021). Precisely, we estimate the treatment effects based on different sub-samples, and calculate the average based on a specific strategy. Furthermore, we implement the strategy from Goodman-Bacon (2021) who claims the estimate by TWFE is the weighted average of all two-period DID estimates presented by the data. Therefore, we apply Bacon Decomposition to the balanced panel in our data into four groups including a "never-taker" group. The estimates from these two strategies are shown in column (1) and (2) of Table 6, and both of them are robust.

Considering that the policy may have spillover effects, which could cause the control group near the treatment group to also be affected by a series of impacts brought by the policy, we implement the spillover-robust DID method by Clarke (2017) and introduces neighboring treatment groups into Model (11) as follows:

$$Y_{ijt} = \beta_0 + \beta_1 TV_{ij} \times Post_{jt} + \beta_2 Close_{ijt} + \theta_i + \tau_j + \mu_t + \gamma X_{ijt} + \epsilon_{ijt} \tag{11}$$

where $Close_{ijt}$ is a dummy variable indicating whether the distance from a given family i in county j to the nearest treatment group within a given time period t exceeds the optimal distance. By introducing neighboring treatment groups, the traditional DID assumption of SUTVA (the policy or treatment applied to individual A only affects A's outcome) is relaxed to estimate the treatment effects and the close to treatment effects. This model has two major advantages over our baseline specification: First, it allows the policy effects of the treatment group to spill over to the control group, with the spillover effect depending on the distance to the nearest treatment group. Second, the area with spillover effects can be determined by the data-driven optimal bandwidth rather than arbitrary choice. Fig. 4 shows the relation between the root mean square error (RMSE, see more details in Clarke (2017)) of the model and the choice of distance used in constructing neighboring treatment group. Our result indicates that the spillover effect is most significant within a radius of 15 kilometers from the treatment group.[13] Column (4) of Table 6 presents the estimated spillover effect. It suggests that the direction of the spillover effect is consistent with the policy effect and is statistically significant. However, it is worthy noting that 99% of China's counties' areas exceed 225 square kilometers (vs. 15 kilometers as the radius with the most significant spillover effect). Therefore, we argue that the spillover effect of the treatment group will only occur within the county and will not spill over to areas outside the county.[14]

7. Mechanism

What drives the decrease in household income? To examine the mechanisms through which Taobao Villages affect household income, we decompose the household income into four components: wage income, business income, investment income, and income transfer. Business income refers to the profit from doing business such as selling agricultural products on E-commerce platforms. The estimated impacts of Taobao Villages on four different sources of income are listed from column (1) to (4) of Table 7. The business income significantly increases by 29.6% after the emergence of Taobao Village in the county while wage income significantly decreases by 18.3%. Meanwhile, we do not find evidence of significant effects on investment income or income transfer, as shown in column (3) and (4) of Table 7. In summary, we find that the decrease in total household income is driven by the reduction in wage income which is not fully compensated by changes in other sources of income.

An explanation for the decrease in wage income is that outflowing migrant workers who used to work outside for higher wages return to their hometowns after the emergence of Taobao Villages. The E-business clusters associated with Taobao Villages provide more job opportunities, therefore, encouraging more workers engaged in relevant and local industries, particularly for migrant workers. In line with the argument, we find that the employment rate increases about 0.016 after the emergence of Taobao Villages in the county, as shown in column (1) in Table 8. Column (2) and (3) further show that local residents are also less likely to be

[13] We use the center of a county according to the latitude and the longitude to calculate the distance.

[14] To ensure robustness, we also conducted regressions at the prefecture-level city level (cities are composed of counties. If a prefecture-level city has Taobao villages, the explanatory variable is assigned a value of 1; otherwise, it is 0). The results remain robust.

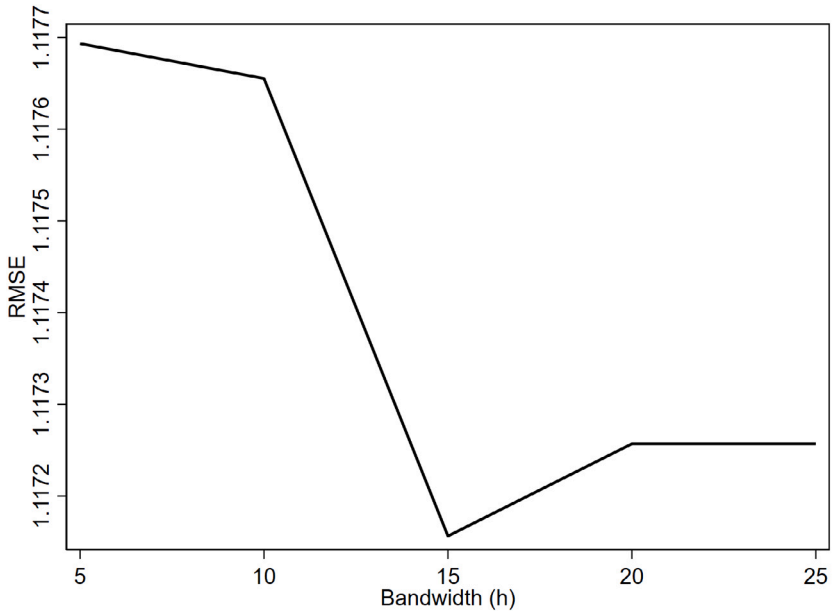

Fig. 4. Optimal distance figure for spillover.

Table 7
The impact of the establishment of Taobao Villages on household income by sources.

	(1) Wage income	(2) Business income	(3) Investment income	(4) Income transfer
$TV_{ij} \times Post_{jt}$	−0.183*	0.296***	0.111	−0.023
	(0.094)	(0.103)	(0.078)	(0.092)
Obs.	55,767	54,591	55,693	54,895

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Control variables in all regressions expect for column (1) include: (1) family level: number of house members, household deposits and urban or rural area (2) house head: age, age squared, sex and education level (3) fixed effects: year fixed effects, family fixed effects, and county fixed effects.

migrant workers and more likely to work in their hometowns. In China, when more household members choose to be local workers rather than migrant workers, the household wage incomes are expected to be lower because the latter group usually earns a much higher wage income (which is the primary motivation to be migrant workers). We attempt to provide some evidence of this. We first focus on the subpopulation that has ever been migrant workers, and column (4) of Table 8 shows that their wage incomes are in fact negatively affected by Taobao Villages. On the contrary, when we focus on the subpopulation who are currently local employed and have never been migrant, we find a significantly positive impact of Taobao Villages on their wage incomes, as shown in column (5). The results indicate that it is indeed opportunity costs that leads to a decrease in wage income.

When Taobao villages encourage people to give up higher wages as migrant workers, they must be rewarded in some way. In this study, we argue that the potential increase in childcare provisions is one type of compensation for the lower income. We split the sample by having left-behind children. Left-behind children refer to children whose parents are migrant workers and do not live with them. Based on the estimates in column (6) and (7), we find a significantly negative effect of Taobao Villages on the number of migrant workers only for households with left-behind children, not for ones without any left-behind children. This result is consistent with the intuition that, given the work opportunities provided in Taobao Villages, households with left-behind children are willing to choose local jobs even with a relatively lower income in exchange for more time spent with their children. Direct evidence is shown in column (8) of Table 8 that dinner time at home increases by about 10% after the emergence of Taobao villages. Columns (9) and (10) of Table 8 provide additional evidence from a gender perspective supporting the motivation to return to family: female workers, who have a stronger desire for family, exhibit a stronger preference for local employment (Koengkan et al., 2022).

The decrease in wage income can also be partially explained by the higher rate of entrepreneurship induced by Taobao Villages. The existence of E-business clusters usually indicates a business-friendly environment with sophisticated infrastructure, logistics services, and social networks, all of which reduce the entry cost of the market. Accordingly, current studies have found that Taobao Villages can encourage local farmers to be entrepreneurs selling their products and services. It is then expected to result in a decline in wage income structurally after more employed workers shift to entrepreneurs. In line with the literature, we find a higher probability

Table 8
The Impact of the establishment of Taobao Villages on household income via decreasing labor income.

Panel A: Impact on labor income					
	(1)	(2)	(3)	(4)	(5)
Dependent Var.	Employed	Migrant worker	Domestic worker	(log) Wage	(log) Wage
Sample				Migrant workers but return	Never been migrant workers
$TV_{ij} \times Post_{jt}$	0.016*	−0.006**	0.021**	−0.255*	0.281**
	(0.009)	(0.003)	(0.009)	(0.152)	(0.123)
P-value				0.000	
Obs.	55,901	55,845	55,901	27,491	5411
Panel B: Motivation for working domestically					
	(6)	(7)	(8)	(9)	(10)
Dependent Var.	Number of migrant persons		(log)Times to eat dinner at home per week	Being migrant worker	
Sample	Have left-behind children	Have no left-behind children		Female	Male
$TV_{ij} \times Post_{jt}$	−0.037**	−0.002	0.101*	−0.009*	0.011
	(0.016)	(0.010)	(0.053)	(0.005)	(0.014)
P-value	0.078			0.091	
Obs.	16,803	39,098	21,934	27,586	27,501

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Control variables in all regressions expect for column (1) include: (1) family level: number of house members, household deposits and urban or rural area (2) house head: age, age squared, sex and education level (3) fixed effects: year fixed effects, family fixed effects, and county fixed effects. Control variables for column (9) and (10) exclude sex.

of being entrepreneurs inspired by Taobao Villages. Columns (4) and (5) of Table 9 report the positive effects of Taobao Villages on the probability of currently running a business and the probability of ever starting one's own business, respectively. The improved access to the Internet and E-commerce is a key component to explain the enlarged population starting businesses. In Panel B of Table 9, we show the impact of the Taobao villages on the use of the Internet. Households significantly increase their Internet use related to work and their business related to the Internet. Meanwhile, the higher rate of participation in business is associated with higher business incomes shown in Table 7.[15] These empirical evidence do not directly support the rationale that the reduction in wage income is caused by the structural switch from wage income to business income for some workers. However, we regard this rationale as a plausible argument because wage income accounts for more than 50% of the total income, and the source of the change in business income is more likely the wage income.

8. Heterogeneity

Conditional business success. Profit from running a business is affected by various factors (Zhang et al., 2025). Even exposed to the same business opportunities brought about by Taobao Villages, people with better social-economic status and environmental conditions are likely to start their businesses successfully, thereby gaining higher business incomes. In this study, we measure social-economic status based on two variables: (1) per capita household income in 2010 and (2) gift expenditure in 2014.[16] Meanwhile, we particularly investigate three environmental factors: (1) whether there is an official community website in 2010, (2) the population mobility rate in 2010, and (3) whether there is any enterprise founded by local residents in 2010. Fig. 5 shows the effects of Taobao Villages on business income using these five indicators. The "low group" refers to those whose level of the given measure is lower than the median, and the "high group" refers to the rest. The results confirm that only " households in the high group" and the group with a more open and entrepreneurial environment significantly gain a higher business income. Higher business income is associated with lower wage income in "higher groups" for most of the measures defined above, indicating the potential structural change in income composition, which are shown in Figs. 6 and 7.

Increasing income inequality. Heterogeneous effects on business income indicate that not all households benefit equally from the business opportunities brought about by Taobao Villages. To better understand how these heterogeneous effects affect local

[15] It is worth noting that business income includes agricultural income and enterprise operating income according to the questionnaire. Alternatively, we check the effect on these two types of business income separately. Shown in columns (1) and (2), we find a significant positive effect on business income from enterprises but not on agricultural income. Meanwhile, we also find a decrease in the engagement in agricultural activities, as shown in column (3).

[16] There is no data of gift expenditure in 2010.

Table 9
The impact of the establishment of Taobao Villages on household income via increasing operating income.

	(1)	(2)	(3)	(4)	(5)
Panel A: Impact on income and entrepreneur					
	(log)Value of agricultural products	(log)Operating income of enterprise	Agricultural engagement	Running private business	Start business
$TV_{ij} \times Post_{jt}$	0.093	0.144**	−0.029***	0.017**	0.021***
	(0.138)	(0.066)	(0.009)	(0.008)	(0.006)
Obs.	51,424	55,901	55,901	55,900	55,901
Panel B: Motivation for entrepreneur					
	Internet for work	Internet for information	Start business and use Internet for work	Start business and use Internet for information	
$TV_{ij} \times Post_{jt}$	0.015***	0.025**	0.003*	0.020***	
	(0.005)	(0.010)	(0.002)	(0.005)	
Obs.	44,721	44,721	44,721	44,721	

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Control variables in all regressions expect for column (1) include: (1) family level: number of house members, household deposits and urban or rural area (2) house head: age, age squared, sex and education level (3) fixed effects: year fixed effects, family fixed effects, and county fixed effects.

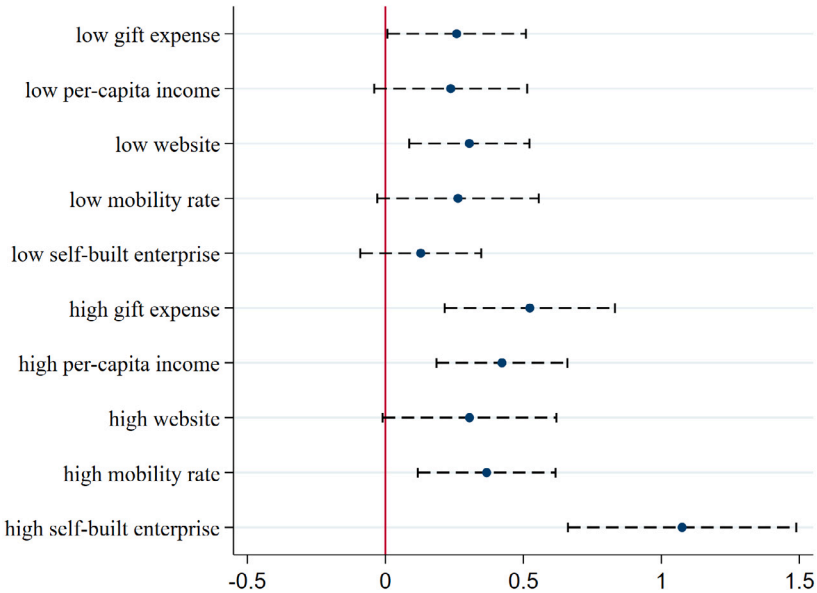

Fig. 5. Heterogeneity in business income.
Note: This figure is derived from the grouped regression, and the values represent the coefficients under different grouping scenarios (for details, see the appendix, which includes the regression coefficients for each group and the inter-group coefficient difference test).

farmers' incomes, we examine the effect of Taobao Villages on the local income inequality. Columns (1) and (2) of Table 10 report the effect of Taobao Villages on Gini coefficient at the county level and the village level, respectively, and the results show that Taobao villages exacerbate income inequality. The increase in income inequality is robust when using the Kakwani index of relative income deprivation as an alternative measure, as shown in column (3) and (4) of Table 10.

9. Conclusion

Based on CFPS data from 2010 to 2020, this article examines the impact of Taobao Villages on the income of local farmers. The findings reveal a significant negative effect of Taobao Villages on local farmers' income. We decompose farmers' income structure and find that the formation of Taobao Villages increases farmers' entrepreneurial income but significantly reduces their wage income,

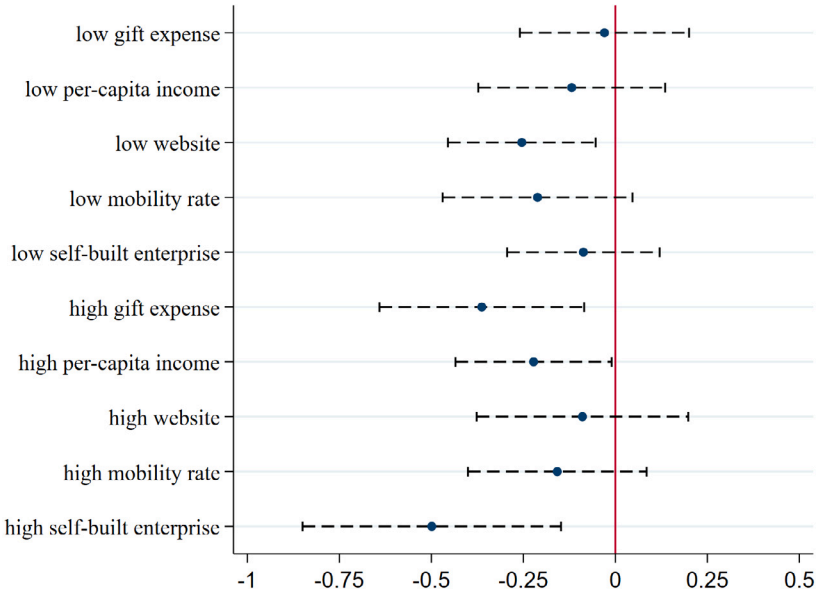

Fig. 6. Heterogeneity in wage income.
This figure is derived from the grouped regression, and the values represent the coefficients under different grouping scenarios (for details, see the appendix, which includes the regression coefficients for each group and the inter-group coefficient difference test).

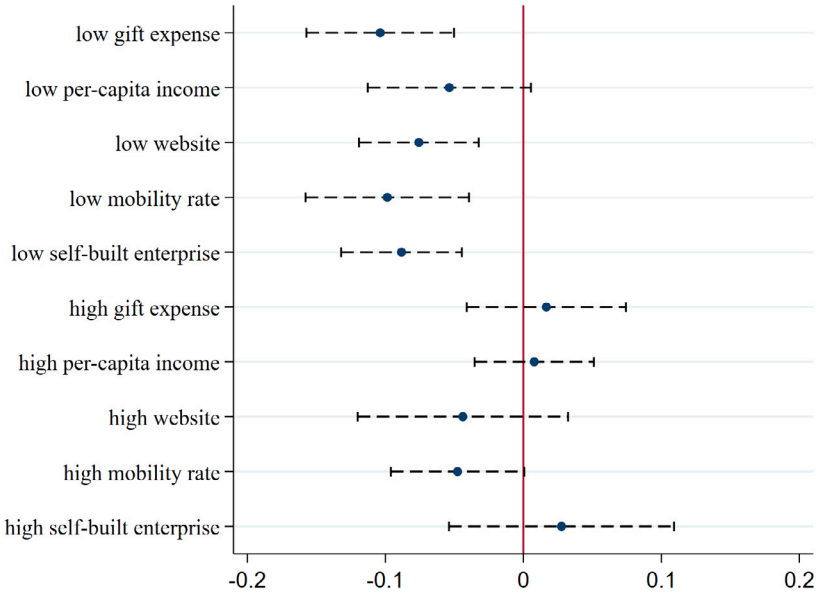

Fig. 7. Heterogeneity in total income.
This figure is derived from the grouped regression, and the values represent the coefficients under different grouping scenarios (for details, see the appendix, which includes the regression coefficients for each group and the inter-group coefficient difference test).

resulting in an overall decline in income. This is attributed to the voluntary abandonment of higher-wage employment opportunities in out-of-town work by local farmers, who choose to return to their hometowns for employment or entrepreneurship, driven by family and childcare-related considerations. Further analysis indicates that Taobao Villages only enable a small proportion of people with higher social and human capital to succeed in entrepreneurship, exacerbating income inequality in the local community.

Our findings suggest that while E-commerce clusters stimulate economic activity, policymakers must address two critical risks: (1) the aggregate decline in household income as stable labor earnings are replaced by volatile business income and (2) the disproportionate burden on disadvantaged households, which risks entrenching systemic inequality. To mitigate the first issue,

Table 10
The effect of Taobao Village on income inequality.

	(1) Gini (county)	(2) Gini (village)	(3) Kakwani index (county)	(4) Kakwani index (village)
$TV_{ij} \times Post_{jt}$	0.022***	0.021***	0.018***	0.018***
	(0.002)	(0.003)	(0.002)	(0.002)
Obs.	53,492	53,059	53,492	53,059

***, **, * refers to significance at 1%, 5%, and 10% significance level respectively. Standard errors are clustered at household level. Control variables in all regressions include: (1) family level: number of house members, household deposits, and urban or rural area (2) house head: age, age squared, sex and education level (3) fixed effects: year fixed effects, family fixed effects, and county fixed effects.

policymakers should prioritize monitoring structural income shifts through granular household surveys, such as Indonesia's National Socioeconomic Survey, which explicitly tracks E-commerce income to inform adjustments to safety nets or financial inclusion programs.

To combat inequality, interventions must directly address the barriers faced by disadvantaged groups. For example, subsidized platform fees for rural households reduce upfront costs that disproportionately discourage low-income families from participating in E-commerce. Similarly, expanding broadband infrastructure (e.g. China's rural digitalization programs) and skill-building initiatives (e.g. Kenya's Digital Literacy Program) can narrow the 'digital divide' by equipping marginalized groups with tools to compete in online markets. Critically, these efforts should be paired with longitudinal evaluations to ensure they reduce, rather than reproduce, existing disparities. By targeting the roots of inequality (access, skills, and cost barriers), policy makers can ensure E-commerce growth fosters inclusive prosperity.

Our study has several limitations that merit consideration. First, our analysis focuses on short-term impacts (within three years), which may not capture longer-term dynamics such as household adaptation to income volatility or delayed benefits of digital skill-building programs. Future research could extend the time frame to 5–10 years to assess whether initial trends stabilize or reverse. Second, the geographic specificity of our sample — Taobao Villages in China — limits generalizability to regions with differing digital infrastructure or governance models. Replicating this work in contexts such as India's Digital India initiative or Africa's platform ecosystems (for example, Jumia) could clarify how institutional and socioeconomic contexts shape results.

CRediT authorship contribution statement

Li Qiuyun: Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mingjia Xie:** Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Yuehuan Tang:** Methodology, Conceptualization.

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